

Comparitive Performance Report for tentacle-ec-4+2-16k-balance-reads-off- run2 vs direct-reads-pr2-ec-4+2-16k-balance-off vs direct-reads-pr2-ec-4+2-16k-balance-on- run2

Table of contents

- Comparison summary for tentacle-ec-4+2-16k-balance-reads-off-run2 vs direct-reads-pr2-ec-4+2-16k-balance-off vs direct-reads-pr2-ec-4+2-16k-balance-on-run2
- Response Curves
 - Sequential Read
 - Random Read
 - Random Read/Write
- Configuration yaml files
 - results

Comparison summary for tentacle-ec-4+2-16k-balance-reads-off-run2 vs direct-reads-pr2-ec-4+2-16k-balance-off vs direct-reads-pr2-ec-4+2-16k-balance-on-run2

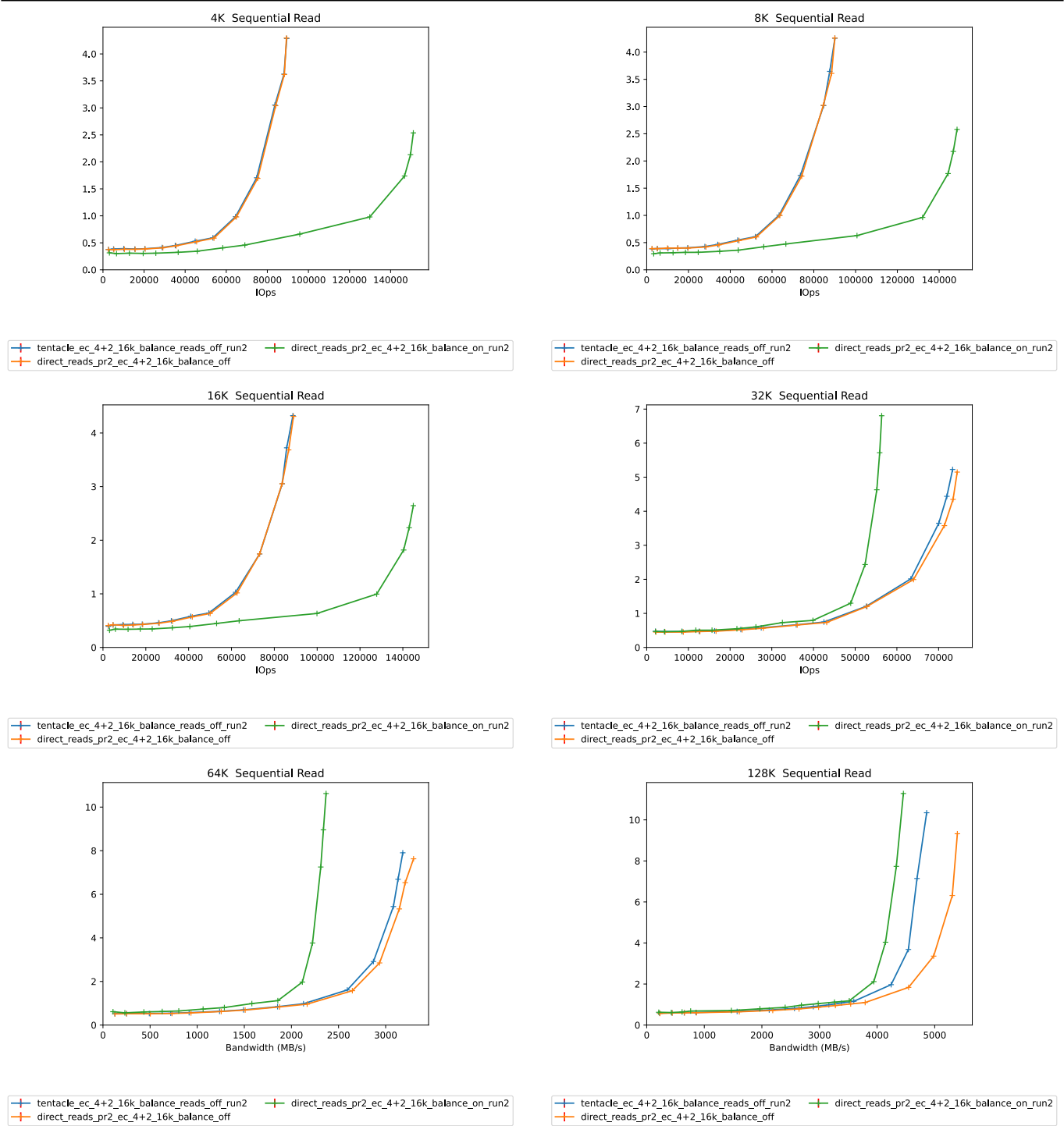
Sequential Read	tentacle_ec_4+2_16k-balance-reads-off-run2	direct-reads-pr2-ec-4+2-16k-balance-off	change	direct-reads-pr2-ec_4+2_16k-balance-on-run2	change
4K	89458@4.3ms	89308@4.3	-0%	151063@2.5	69%
8K	90112@4.3ms	90051@4.3	-0%	148569@2.6	65%
16K	88672@4.3ms	88958@4.3	0%	144892@2.6	63%
32K	73365@5.2ms	74468@5.1	2%	56357@6.8	-23%
64K	3182@7.9ms	3296@7.6	4%	2368@10.6	-26%
128K	4861@10.3ms	5394@9.3	11%	4457@11.3	-8%
256K	6711@15.0ms	7239@9.3	8%	7482@13.4	11%
384K	6865@29.3ms	8059@25.0	17%	8750@23.0	27%
512K	7275@36.9ms	8712@30.8	20%	10333@13.0	42%
768K	7488@26.9ms	8867@45.4	18%	11093@9.1	48%
1024K	8047@16.7ms	9069@44.4	13%	12541@21.4	56%
2048K	8274@48.7ms	9018@44.7	9%	13579@9.9	64%
4096K	7953@16.9ms	8673@61.9	9%	12707@10.5	60%

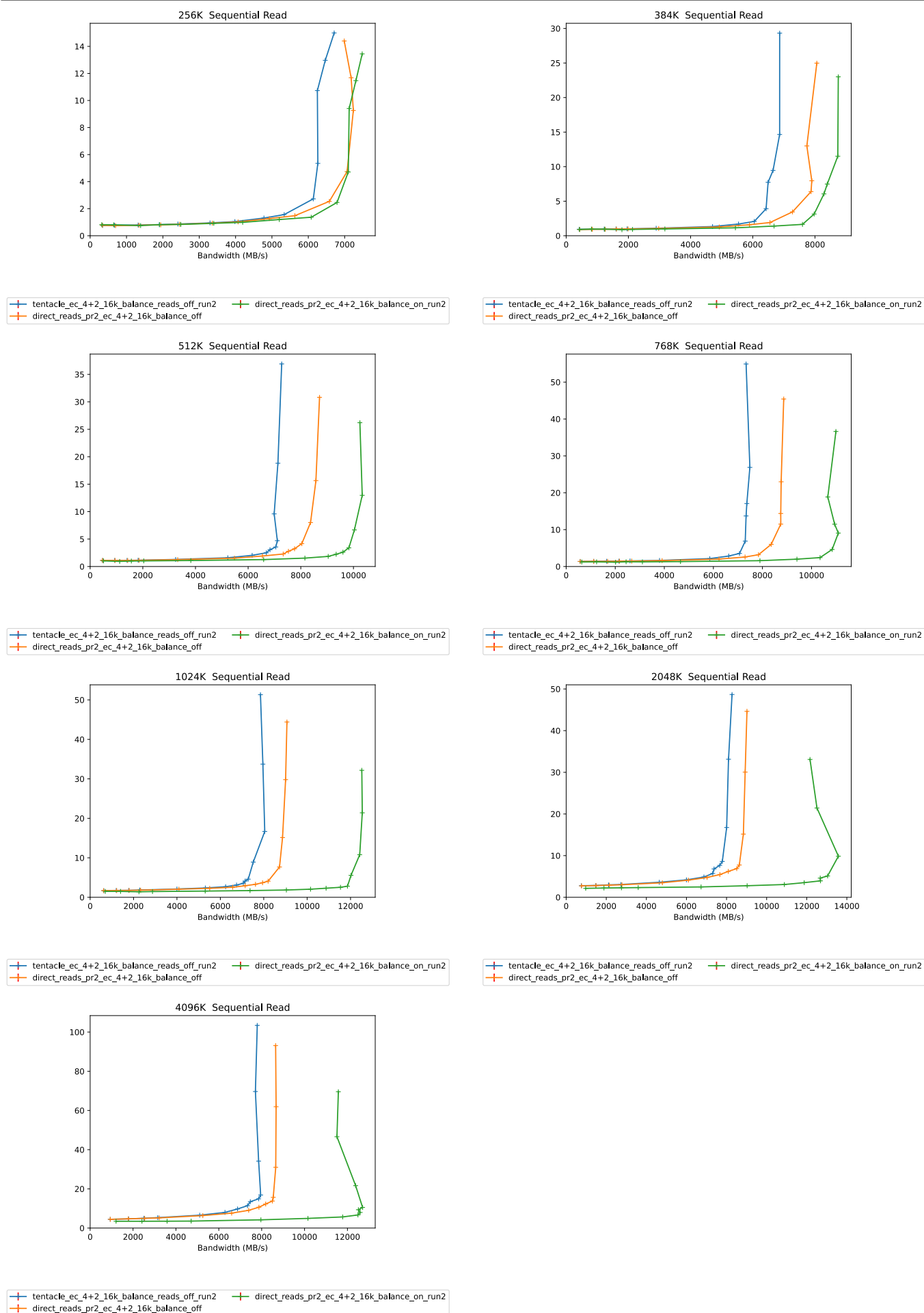
Random Read	tentacle_ec_4+2_16k-balance-reads-off-run2	direct-reads-pr2-ec-4+2-16k-balance-off	change	direct-reads-pr2-ec_4+2_16k-balance-on-run2	change
4K	109637@3.5ms	112826@3.4	3%	147506@2.6	35%
8K	105499@3.6ms	108129@3.5	2%	145356@2.6	38%
16K	100733@3.8ms	101958@3.8	1%	139453@2.7	38%
32K	68357@4.7ms	71338@4.5	4%	59527@6.4	-13%
64K	2780@6.0ms	2930@7.2	5%	2350@10.7	-15%
128K	4613@7.3ms	4975@10.1	8%	4216@11.9	-9%
256K	6111@5.5ms	6840@4.9	12%	6922@12.1	13%
384K	6789@3.7ms	7644@13.2	13%	8486@23.7	25%
512K	6933@19.4ms	8193@32.8	18%	9894@27.1	43%
768K	7598@53.0ms	8678@46.4	14%	10690@18.8	41%
1024K	7620@8.8ms	8938@30.0	17%	11855@34.0	56%
2048K	8553@47.1ms	9101@29.5	6%	13221@10.1	55%
4096K	8481@31.6ms	8731@30.7	3%	13557@19.8	60%

Random Read/Write	tentacle_ec_4+2_16k-balance-reads-off-run2	direct-reads-pr2-ec-4+2-16k-balance-off	change	direct-reads-pr2-ec_4+2_16k-balance-on-run2	change
4K_70/30	46104@8.3ms	49029@6.5	6%	53288@6.0	16%
16K_70/30	42227@9.1ms	46327@8.3	10%	49888@7.7	18%
64K_30/70	1303@19.3ms	1394@18.0	7%	1305@19.3	0%
64K_70/30	1741@12.0ms	1822@11.5	5%	1617@15.5	-7%

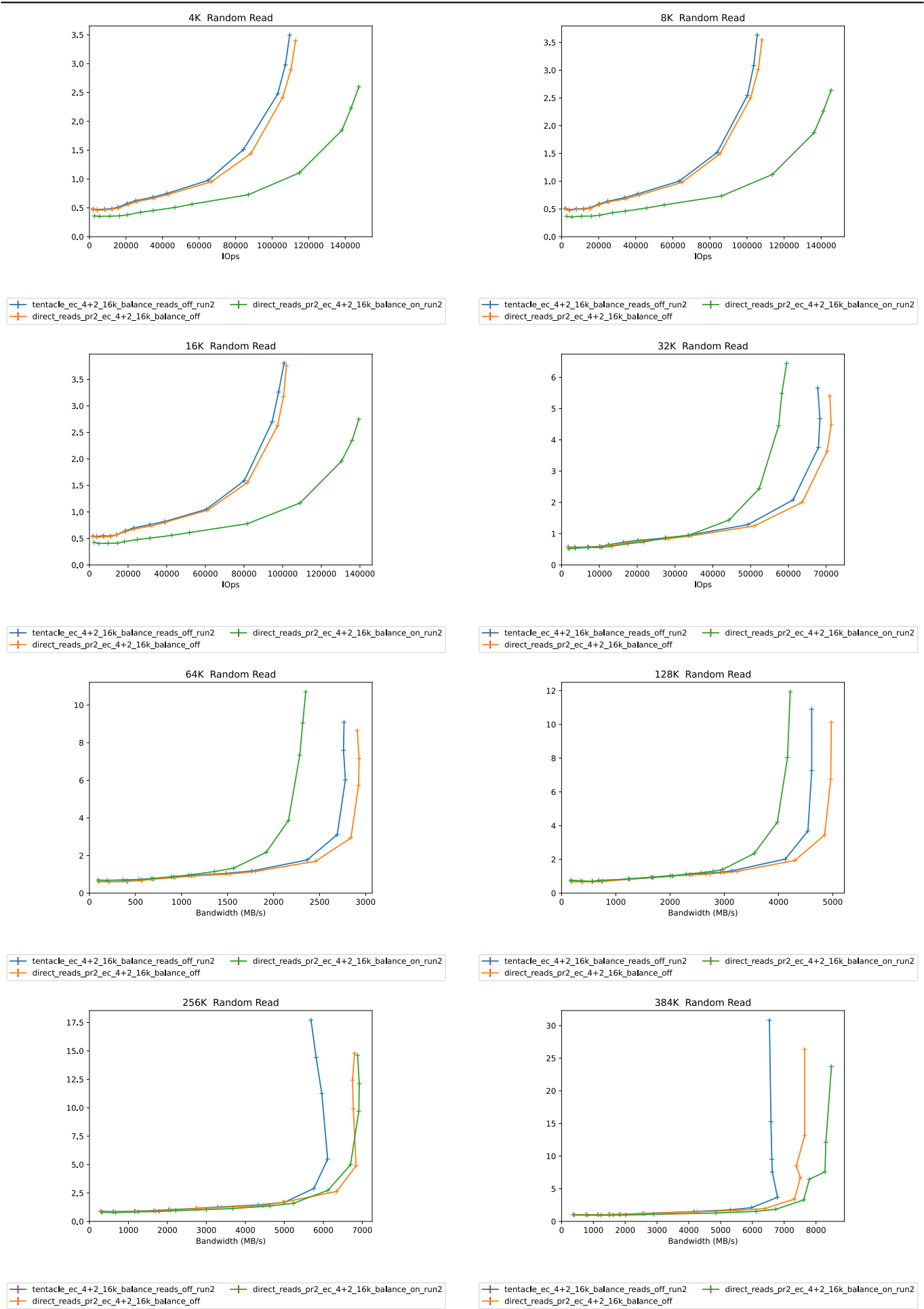
Response Curves

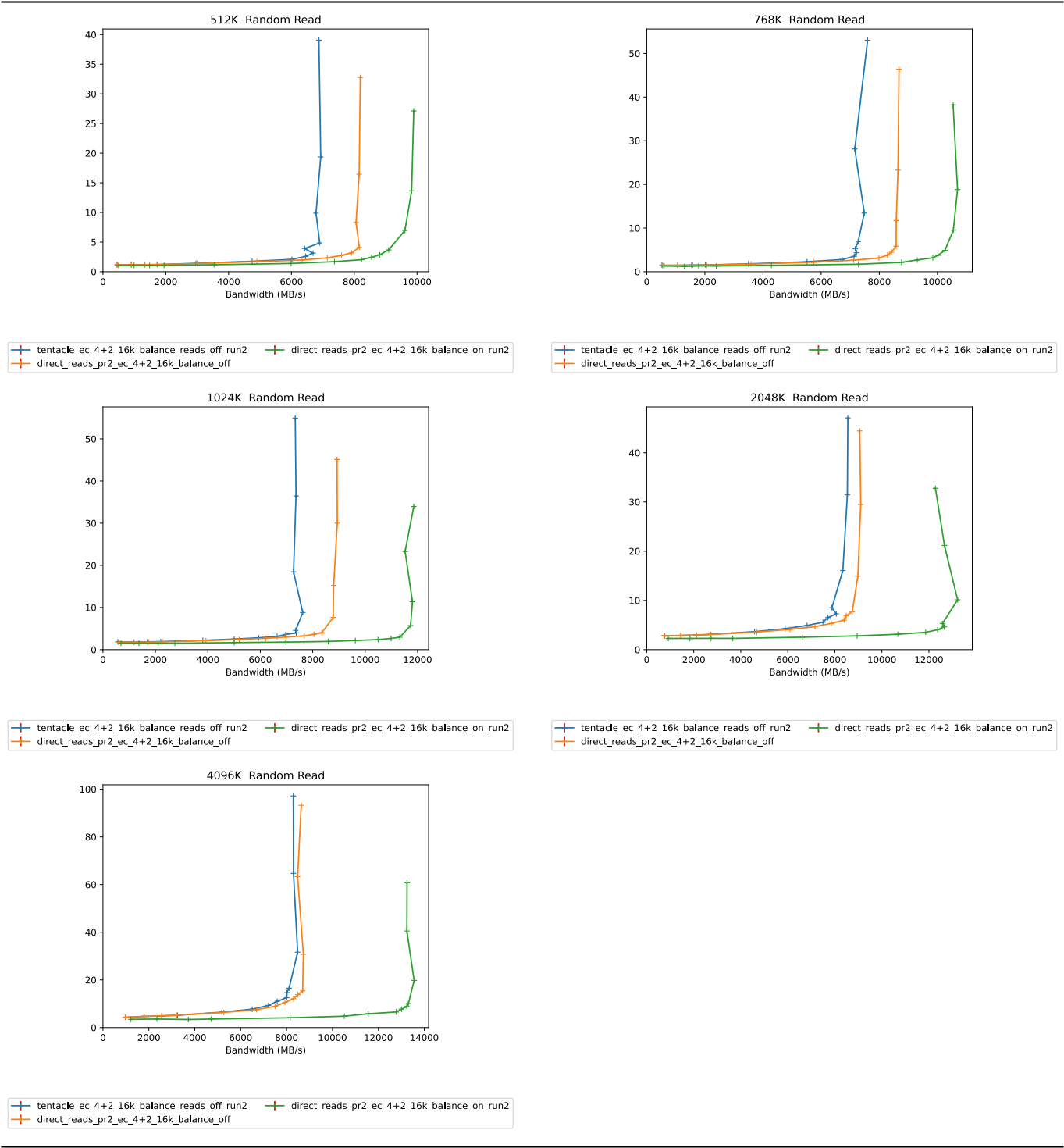
Sequential Read



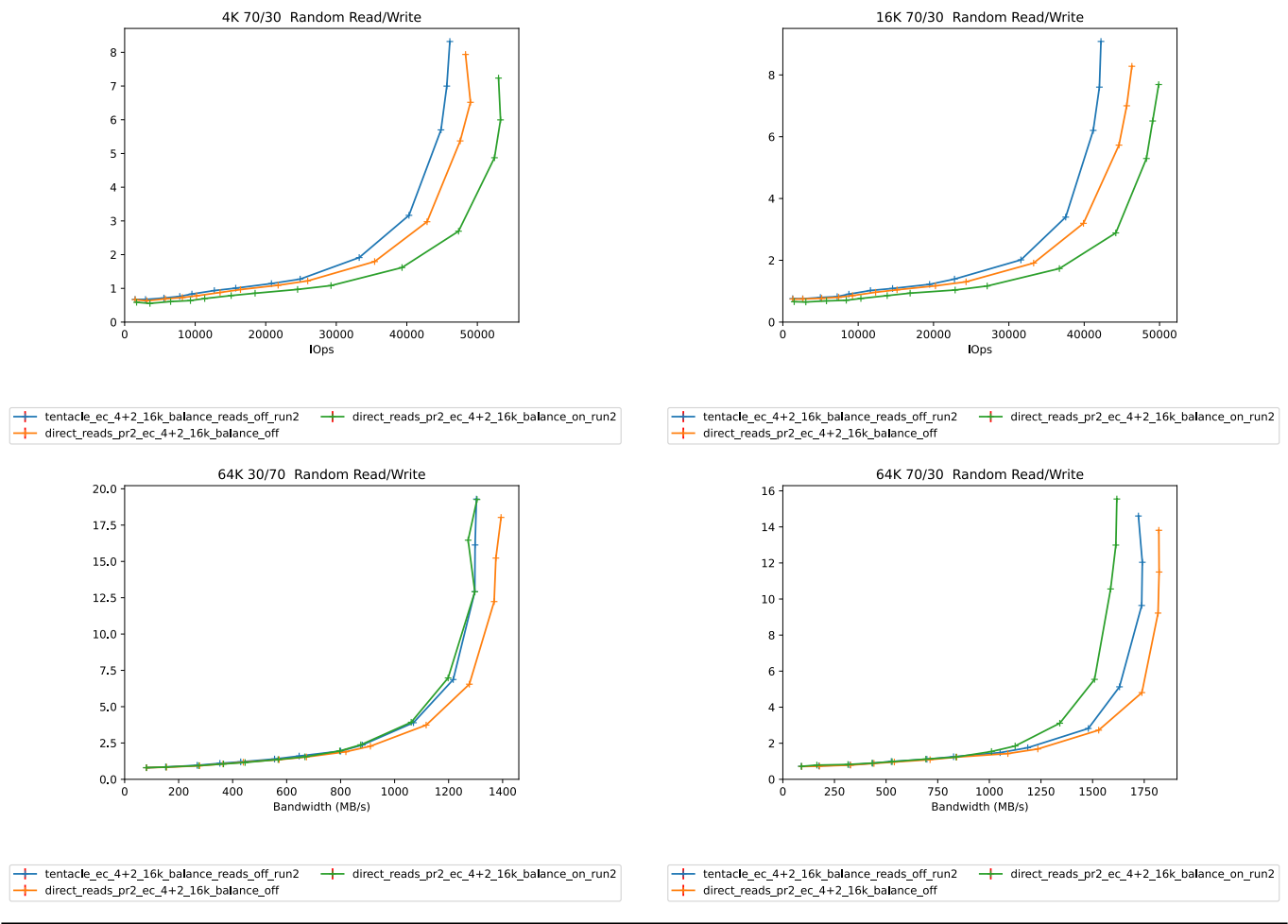


Random Read





Random Read/Write



Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
4kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:

```

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64kmixread70:
    jobname: mixread
    mode: randrw
    numjobs:
      - 1
    op_size: 65536
    rwmixread: 70
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1

```

```

    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  768krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 786432
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 40
      - 48
      - 64
      - 128
      - 256
      - 512
  768ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 786432
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 5
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 160
      - 256
      - 512
  8krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}
```